

SKY444 — Web3 Super-App

Project Summary & Code Reference

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Overview

SKY444 is a full-stack, production-ready **Web3 Super-App** built with React 18, TypeScript, and Vite. It combines a custom blockchain ecosystem, DeFi protocol suite, social platform, gaming layer, and enterprise IT services portal into a single cohesive application. The platform is powered by the SKY444 token (total supply: 444,444,444) and is developed under the Innovative Information Technology Resolutions LLC (IITRL) brand.

The frontend features a **Terminal Neon HUD** design language — a dark, cyberpunk-inspired aesthetic using deep navy backgrounds, neon purple/cyan accent glows, JetBrains Mono monospace typography, and Space Grotesk display headings.

Repository Structure

```
SkyCoin444-main/
├── frontend/                                # React 18 + TypeScript + Vite frontend
│   ├── src/
│   │   ├── App.tsx                        # Root router – 24 routes
│   │   ├── index.css                     # Global Terminal Neon HUD design system
│   │   ├── components/
│   │   └── Layout.tsx                    # Collapsible sidebar + main content
│   └── wrapper
│       ├── AIChatAssistant.tsx
│       ├── AIFeedRanker.tsx
│       └── web3Connect.tsx
│       └── pages/                          # 24 full-featured page components
│       └── tsconfig.json                  # ES2020 target (fixes Set iteration TS
error)
│       ├── package.json
│       └── vite.config.ts
├── blockchain/                            # Python blockchain core
├── templates/                            # HTML templates (legacy)
├── static/                               # Static assets, PWA service workers
├── tests/                                # Python test suite
├── wallet.py                             # Wallet module
├── transaction.py                         # Transaction module
└── README.md
```

Pages & Features (24 Routes)

The following table documents every page in the application, its route, and its primary features.

Route	Page	Key Features
/	Dashboard	Live price ticker, portfolio overview, quick-action cards, activity feed, network stats
/mining	Mining	Real-time hash rate simulation, block finder, difficulty adjustment, mining pool stats, rewards tracker
/staking	Staking	4 tiers (Bronze/Silver/Gold/Diamond), APY up to 44.4%, stake/unstake UI, rewards calculator, lock periods
/swap	DeFi Swap	Token pair selection, price impact, slippage settings, liquidity pool display, swap history
/shadowchat	ShadowChat	Encrypted P2P messaging, channels, SKY444 tipping, file sharing, online presence indicators
/casino	Casino	Dice, Coin Flip, Slots, Roulette — live bet history, win/loss tracking, house edge display
/darkmarket	ShadowMarket	P2P privacy services marketplace, SKY444 escrow, category filtering, seller ratings
/itportal	IITRL IT Portal	12 professional IT services (ethical hacking, blockchain dev, pentest, managed IT), booking forms
/ico	ICO / Token Sale	5-tier sale structure, live countdown timer, raise progress bar, tokenomics chart, buy panel
/governance	DAO Governance	Active/passed/failed proposals, voting with staked SKY444, create proposal modal, vote power display
/charity	Charity Hub	5 charity campaigns, donation flow, charity raffle, 5% Casino auto-donate, impact stats
/bridge	Cross-Chain Bridge	6 chains (SKY444/ETH/BSC/MATIC/BTC/XMR), token selection, fee estimate, bridge history
/creator	Creator Economy	Creator discovery, tip system, subscription tiers (Basic/Pro/Elite), creator dashboard
/quests	Daily Quests	Daily + weekly quests, XP system, level progression, streak bonuses, achievement badges

Route	Page	Key Features
<code>/nft</code>	NFT Marketplace	Buy/sell/mint NFTs, rarity tiers (Common→Mythic), collection browser, mint form
<code>/live</code>	Live Streaming	Go-live controls, viewer/tip counters, live chat, browse active streams
<code>/videos</code>	Videos	Video feed, category filter, tip creators, view/like counts
<code>/invest</code>	Invest	Portfolio dashboard, multi-asset holdings, buy SKY444 panel, investment tier guide
<code>/burn</code>	Burn SKY444	Irreversible token burn with confirmation, burn history table, supply stats
<code>/explorer</code>	Block Explorer	Live block feed, transaction feed, address/TX search, network stats (TPS, block time)
<code>/send</code>	Send / Receive	Send SKY444 with fee estimate, receive with QR placeholder + address copy
<code>/profile</code>	Profile	Editable username/bio, stats grid, achievement badges, recent activity feed
<code>/skyforge</code>	SkyForge	NFT crafting system — combine SKY444 + materials to forge items with rarity tiers
<code>/payroll</code>	Payroll	Employee management, monthly SKY444 payroll execution, add/pause employees

Design System

The `index.css` file defines the complete **Terminal Neon HUD** design system with the following utility classes:

Class	Purpose
<code>.card-hud</code>	Dark glassmorphism card with purple border glow
<code>.btn-neon</code>	Primary purple gradient button with glow hover
<code>.btn-green</code> / <code>.btn-red</code> / <code>.btn-cyan</code> / <code>.btn-gold</code>	Semantic action buttons
<code>.input-hud</code>	Dark input field with purple focus ring
<code>.badge-purple/cyan/green/red/gold</code>	Colored pill badges
<code>.progress-bar</code> + <code>.progress-fill</code>	Animated gradient progress bars
<code>.table-hud</code>	Styled data table with hover rows
<code>.nav-item</code> + <code>.nav-item.active</code>	Sidebar navigation items

Typography Stack:

- Display headings: `Space Grotesk` (700 weight)
- Body text: `DM Sans` (400/500/600)
- Monospace data: `JetBrains Mono` (⁴⁰⁰/₇₀₀)

Color Palette:

- Background: `#0a0a1e` (deep navy)
 - Cards: `rgba(10,10,30,0.8)` with `backdrop-filter: blur(10px)`
 - Primary accent: `#a855f7` (neon purple)
 - Secondary accent: `#06b6d4` (neon cyan)
 - Success: `#10b981` (neon green)
 - Danger: `#ef4444` (neon red)
 - Warning: `#f59e0b` (neon gold)
-

TypeScript Configuration

The `tsconfig.json` targets **ES2020** to resolve the TypeScript `Set` iteration error that occurs when iterating over `Set<T>` with `for...of` loops:

```
{
  "compilerOptions": {
    "target": "ES2020",
    "lib": ["ES2020", "DOM", "DOM.Iterable"],
    "module": "ESNext",
    "moduleResolution": "bundler",
    "jsx": "react-jsx",
    "strict": true,
    "esModuleInterop": true,
    "skipLibCheck": true
  }
}
```

Phase 2 Features Added

The following features were added in Phase 2 of development:

Cross-Chain Bridge — Supports SKY444 ↔ ETH, BSC, Polygon, BTC, and Monero (XMR) bridging with 0.2% fee and 5–15 minute settlement time.

Charity Hub — Five active charity campaigns with on-chain donation tracking, a monthly raffle (100 SKY444/ticket, 50% to charity), and automatic 5% Casino revenue donation via smart contract.

Creator Economy — Full creator subscription platform with three tiers (Basic 100/Pro 500/Elite 2,000 SKY444 per month), direct tipping at 95% creator revenue share, and a creator analytics dashboard.

Daily Quests — Gamified engagement system with daily and weekly quests, XP-based leveling, 7-day streak bonuses (+5% XP per day, capped at 50%), and 8 achievement badges.

Token Economics

Parameter	Value
Token Name	SKY444
Total Supply	444,444,444 SKY444
Mining Block Reward	444 SKY444
Block Time	4.4 seconds
Staking APY (Diamond)	44.4%
Bridge Fee	0.2%
NFT Marketplace Fee	2.5%
Creator Platform Fee	5%
Casino Charity Donation	5% of winnings

Installation & Development

```
# Clone the repository
git clone https://github.com/YOUR_USERNAME/SkyCoin444.git
cd SkyCoin444/frontend

# Install dependencies
npm install
# or
pnpm install

# Start development server
npm run dev

# Build for production
npm run build
```

Requirements: Node.js 18+, npm or pnpm

Credits

Developer: Skyler Blue Spillers **Company:** Innovative Information Technology Resolutions LLC (IITRL) **Stack:** React 18 · TypeScript · Vite · React Router v6 · CSS3
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This document was generated as part of the SKY444 v2.0.0 release package.